SAT Math Mastery · Session 3

Nonlinear Equations and Systems

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics solving quadratics: factoring, completing the square, discriminant, sum and product of roots · line and parabola, line and circle systems · number of solutions and unknown constants · rational equations and extraneous solutions · exponential equations · absolute value, radical and higher-degree equations

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. Medium · No calculator needed

$$\frac{1}{x^2+10x+25}=4$$

Which of the following could be the value of $x+5$?

(A) $\frac{1}{2}$ ✓ (B) $\frac{5}{2}$

(C) $\frac{9}{2}$ (D) $\frac{11}{2}$

1. The denominator is a perfect square: $x^2+10x+25=(x+5)^2$. So the equation says $\frac{1}{(x+5)^2}=4$.

2. Flip both sides (take the reciprocal): $(x+5)^2=\frac{1}{4}$.

3. Take the square root of both sides: $x+5=\pm\frac{1}{2}$, so $x+5$ could be $\frac{1}{2}$ or $-\frac{1}{2}$.

4. Only $\frac{1}{2}$ is listed, so the answer is (A). Careful: solving for x gives $-\frac{9}{2}$ or $-\frac{11}{2}$, but the question asks for $x+5$.

Answer: (A)

2. Hard · No calculator needed

In the xy-plane, the graphs of $y=x^2-6x+8$ and $y=2x-4$ intersect at two points. What is the sum of the x-coordinates of these two points?

1. Set the two expressions equal: $x^2-6x+8=2x-4$.

2. Move everything to the left side: $x^2-8x+12=0$.

3. Factor: $(x-2)(x-6)=0$, so the two points have $x=2$ and $x=6$.

4. The sum is $2+6=8$. (Shortcut: the roots of $x^2+bx+c=0$ always add up to $-b$, and here $-(-8)=8$.)

Answer: 8

3. HARD · NO CALCULATOR NEEDED

In the xy-plane, the graphs of $y = x^2 - 4x - 5$ and $y = 3x + 1$ intersect at two points. What is the product of the x-coordinates of these two points?

1. Set the two expressions equal: $x^2 - 4x - 5 = 3x + 1$.

2. Move everything to the left side: $x^2 - 7x - 6 = 0$. This does not factor nicely, and it does not need to.

3. For $x^2 + bx + c = 0$ the roots multiply to c (Vieta's formulas). Here the roots multiply to -6.

4. Check: the roots are $\dfrac{7 \pm \sqrt{73}}{2}$, and their product is $\dfrac{49 - 73}{4} = -6$.

Answer: -6

4. HARD · NO CALCULATOR NEEDED

In the xy-plane, the graphs of $y = x^2 - x - 14$ and $y = 2x - 4$ intersect at two points. What is the positive difference between the x-coordinates of these two points?

1. Set the two expressions equal: $x^2 - x - 14 = 2x - 4$.

2. Move everything to the left side: $x^2 - 3x - 10 = 0$.

3. Factor: $(x + 2)(x - 5) = 0$, so $x = -2$ and $x = 5$.

4. The positive difference is $5 - (-2) = 7$.

Answer: 7

5. HARD · NO CALCULATOR NEEDED

Suppose m and n are positive integers such that $mn > 20$ and $12mn$ is a perfect square. What is the least possible value of mn?

1. Split off the square: $12mn = 4 \cdot 3mn$. Since 4 is already a perfect square, $12mn$ is a perfect square exactly when $3mn$ is.

2. $3mn$ is a perfect square only when $mn = 3k^2$ for some positive integer k. The possible values of mn are $3, 12, 27, 48, \ldots$

3. We need $mn > 20$, and the first value on the list above 20 is 27.

4. Check: $m = 3$ and $n = 9$ give $mn = 27$ and $12mn = 324 = 18^2$. The value before it, 12, is not above 20. The answer is 27.

Answer: 27

6. HARD · NO CALCULATOR NEEDED

If $x^2 - 8x + 5 = 0$ has roots r and s, what is the value of $r^2 + s^2$?

1. For $x^2 + bx + c = 0$ with roots r and s: $r + s = -b$ and $rs = c$. Here $r + s = 8$ and $rs = 5$.

2. Rewrite what we need using these: $r^2 + s^2 = (r + s)^2 - 2rs$.

3. Substitute: $8^2 - 2 \cdot 5 = 64 - 10 = 54$.

4. The roots are $4 \pm \sqrt{11}$ (not needed). Check: $(4 + \sqrt{11})^2 + (4 - \sqrt{11})^2 = 2(16 + 11) = 54$.

Answer: 54

7. HARD · NO CALCULATOR NEEDED

What is the positive solution to the equation $x^2 - 6x - 7 = 0$, found by completing the square?

1. Move the constant: $x^2 - 6x = 7$.

2. Take half of -6, which is -3, and square it: 9. Add 9 to both sides: $x^2 - 6x + 9 = 16$.

3. The left side is a perfect square: $(x - 3)^2 = 16$, so $x - 3 = \pm 4$.

4. Then $x = 3 \pm 4$, giving $x = 7$ or $x = -1$. The positive solution is 7.

Answer: 7

8. HARD · NO CALCULATOR NEEDED

If $y = x^2 - 7$ and $x + y = 5$, what is the value of x^2y^2 for the solution (x, y) of this system with $x > 0$?

1. Replace y in $x + y = 5$ with $x^2 - 7$: $x + x^2 - 7 = 5$.

2. Rearrange: $x^2 + x - 12 = 0$, which factors as $(x + 4)(x - 3) = 0$, so $x = -4$ or $x = 3$.

3. We need $x > 0$, so $x = 3$ and $y = 3^2 - 7 = 2$.

4. Then $x^2y^2 = 3^2 \cdot 2^2 = 9 \cdot 4 = 36$.

Answer: 36

9. HARD · NO CALCULATOR NEEDED

A taxi company uses the formula $10p = q + 25r$, where p is the fare per passenger, in dollars; q is a fixed service charge, in dollars; and r is the distance traveled, in kilometers. Which equation correctly expresses p in terms of q and r?

(A) $p = \dfrac{q}{10} + 25r$

(B) $p = q + \dfrac{25r}{10}$

(C) $p = 10\,(q + 25r)$

(D) $p = \dfrac{q + 25r}{10}$ ✓

1. We need p alone. p is multiplied by 10, so divide BOTH sides by 10: $p = \dfrac{q + 25r}{10}$.

2. The division applies to the whole right side, not just one term. That is why $p = \dfrac{q}{10} + 25r$ and $p = q + \dfrac{25r}{10}$ are wrong. Multiplying by 10 instead of dividing is also wrong.

3. Test with numbers: $q = 20$ and $r = 4$ give $10p = 20 + 100 = 120$, so $p = 12$.

4. Only $\dfrac{q + 25r}{10} = \dfrac{120}{10} = 12$ matches. The others give $\dfrac{q}{10} + 25r = 102$, $q + \dfrac{25r}{10} = 30$ and $10(q + 25r) = 1200$. The answer is (D).

Answer: (D)

10. HARD · NO CALCULATOR NEEDED

A company's monthly profit, in thousands of dollars, is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the price of the product in dollars. For what price x, other than $x = 5$, does the company break even (that is, $P(x) = 0$)?

1. Break even means $P(x) = 0$: $-2x^2 + 40x - 150 = 0$.

2. Divide every term by -2: $x^2 - 20x + 75 = 0$.

3. Factor: $(x - 5)(x - 15) = 0$, so $x = 5$ or $x = 15$.

4. The problem says $x = 5$ is already known, so the other price is 15. Check: $P(15) = -2(15)^2 + 40(15) - 150 = 0$.

Answer: 15

11. HARD · NO CALCULATOR NEEDED

A system of equations is given below, where x represents the number of years since a company's founding and y represents its annual revenue, in millions of dollars.

$$y = x^2 - 5x + 20$$

$$y = 3x + 8$$

For how many values of x, with $x \geq 0$, are the two models equal?

(A) 0 (B) 1

(C) 2 ✓ (D) 3

1. The models are equal where the two expressions are equal: $x^2 - 5x + 20 = 3x + 8$.

2. Move everything to the left side: $x^2 - 8x + 12 = 0$.

3. Factor: $(x - 2)(x - 6) = 0$, so $x = 2$ or $x = 6$.

4. Both values satisfy $x \geq 0$, so there are 2 values. The answer is (C).

Answer: (C)

12. HARD · NO CALCULATOR NEEDED

$$x^2\,(x + 5)\,(x - b) = 0$$

In the given equation, b is a positive constant. The sum of the solutions of the equation is 7. What is the value of b?

1. Read off the solutions: $x^2 = 0$ gives $x = 0$, $x + 5 = 0$ gives $x = -5$, and $x - b = 0$ gives $x = b$.

2. Even though the factor x^2 is squared, the solution $x = 0$ is counted only once. The distinct solutions are 0, -5 and b.

3. Their sum is $0 + (-5) + b = 7$, so $b - 5 = 7$.

4. Add 5: $b = 12$. This is positive, as the problem requires.

Answer: 12

13. HARD · NO CALCULATOR NEEDED

If the graphs of $y = x^2 + 1$ and $y = -5x + p$ intersect at the point $(2, r)$, what is the value of $\frac{p}{r}$?

1. The point $(2, r)$ is on $y = x^2 + 1$, so $r = 2^2 + 1 = 5$.
2. The same point is on $y = -5x + p$, so $5 = -5(2) + p = -10 + p$.
3. Add 10: $p = 15$.
4. Then $\frac{p}{r} = \frac{15}{5} = 3$.
Answer: 3

14. HARD · NO CALCULATOR NEEDED

The functions f and h are defined by $f(x) = x^2 + 3x - 40$ and $h(x) = x^2$. If $3h(x) = f(2x)$, what is the positive value of x?
1. First build $f(2x)$ by replacing every x in f with $2x$: $f(2x) = (2x)^2 + 3(2x) - 40 = 4x^2 + 6x - 40$.
2. Set $3h(x) = f(2x)$: $3x^2 = 4x^2 + 6x - 40$.
3. Move everything to the right side: $0 = x^2 + 6x - 40$.
4. Factor: $(x + 10)(x - 4) = 0$, so $x = -10$ or $x = 4$. The positive value is 4.
Answer: 4

15. HARD · NO CALCULATOR NEEDED

Let $f(x) = ax^3$ and $g(x) = bx^5$ for any value of x. If a and b are positive constants, for how many values of x is $f(x) = g(x)$?

(A) None (B) One

(C) Two **(D) Three ✓**

1. Move everything to one side: $bx^5 - ax^3 = 0$.
2. Factor out x^3: $x^3(bx^2 - a) = 0$. So either $x^3 = 0$ or $bx^2 = a$.
3. $x^3 = 0$ gives $x = 0$. And $x^2 = \frac{a}{b}$ is positive (both a and b are positive), so it gives two more values, $x = \pm\sqrt{\frac{a}{b}}$.
4. Example: $a = b = 1$ gives $x^3 = x^5$ at $x = -1$, 0, 1, which is 3 values. The answer is (D).
Answer: (D)

16. HARD · NO CALCULATOR NEEDED

If $(x + 3)^2 = 1$ and $(x - 4)^2 = 36$, what is the value of x?
1. Solve the first equation: $x + 3 = \pm 1$, so $x = -2$ or $x = -4$.
2. Solve the second equation: $x - 4 = \pm 6$, so $x = 10$ or $x = -2$.
3. The value of x must make BOTH equations true, so it is the value that appears in both lists: $x = -2$.
4. Check: $(-2 + 3)^2 = 1$ and $(-2 - 4)^2 = 36$.
Answer: -2

17. HARD · NO CALCULATOR NEEDED

What value of x satisfies the equation below?

$$\frac{x}{x-3}+\frac{2}{x}=\frac{3}{x-3}$$

1. Multiply every term by the common denominator $x(x-3)$: $x\cdot x+2(x-3)=3x$.
2. Expand and move everything to one side: $x^2+2x-6-3x=0$, so $x^2-x-6=0$.
3. Factor: $(x+2)(x-3)=0$, so $x=-2$ or $x=3$.
4. $x=3$ makes the denominator $x-3$ equal to 0, so it is not allowed. The answer is $x=-2$. Check: $\frac{-2}{-5}+\frac{2}{-2}=-\frac{3}{5}$ and $\frac{3}{-5}=-\frac{3}{5}$.

Answer: -2

18. HARD · NO CALCULATOR NEEDED

What value of x satisfies the equation $2^{3x}\cdot 4^{x+1}=8^{x+3}$?

1. Write every power with base 2: $4=2^2$ and $8=2^3$. The equation becomes $2^{3x}\cdot(2^2)^{x+1}=(2^3)^{x+3}$.
2. Use $(a^m)^n=a^{mn}$: $(2^2)^{x+1}=2^{2x+2}$ and $(2^3)^{x+3}=2^{3x+9}$. So $2^{3x}\cdot 2^{2x+2}=2^{3x+9}$.
3. Add exponents when multiplying powers with the same base: $2^{5x+2}=2^{3x+9}$, so $5x+2=3x+9$.
4. Solve: $2x=7$, so $x=\frac{7}{2}$ (or 3.5). Check: the left exponent is $3x+2(x+1)=\frac{39}{2}$ and the right exponent is $3(x+3)=\frac{39}{2}$.

Answer: $\frac{7}{2}$ **(or** 3.5**)**

19. HARD · NO CALCULATOR NEEDED

$$y=x^2-4x+1$$
$$y=2x-8$$

How many solutions does the system above have?

(A) Zero
(B) Exactly one ✓
(C) Exactly two
(D) Infinitely many

1. Set the expressions equal: $x^2-4x+1=2x-8$.
2. Move everything to the left side: $x^2-6x+9=0$.
3. Count the solutions with the discriminant: $b^2-4ac=(-6)^2-4(1)(9)=0$.
4. A discriminant of 0 means exactly one solution. Indeed $x^2-6x+9=(x-3)^2$, so $x=3$ only. The answer is (B).

Answer: (B)

20. HARD · NO CALCULATOR NEEDED

The equation $2x^2 - 12x + k = 0$, where k is a constant, has two real solutions. One solution is twice the other solution. What is the value of k?

1. Call the solutions r and $2r$. For $2x^2 - 12x + k = 0$ the solutions add up to $-\dfrac{b}{a} = \dfrac{12}{2} = 6$.
2. So $r + 2r = 6$, which gives $3r = 6$ and $r = 2$. The solutions are 2 and 4.
3. The solutions multiply to $\dfrac{c}{a} = \dfrac{k}{2}$, so $\dfrac{k}{2} = 2 \cdot 4 = 8$.
4. Then $k = 16$. Check: $2x^2 - 12x + 16 = 2(x - 2)(x - 4)$.

Answer: 16

21. HARD · NO CALCULATOR NEEDED

For a real number x, $4^x - 2^{x+2} - 32 = 0$. What is the value of 2^{x+1}?

(A) -8 (B) 8

(C) 16 ✓ (D) 32

1. Write everything in terms of 2^x: $4^x = (2^x)^2$ and $2^{x+2} = 4 \cdot 2^x$. Let $u = 2^x$, so $u^2 - 4u - 32 = 0$.
2. Factor: $(u + 4)(u - 8) = 0$, so $u = -4$ or $u = 8$.
3. A power of 2 is never negative, so $u = -4$ is rejected and $2^x = 8$.
4. The question asks for $2^{x+1} = 2 \cdot 2^x = 2 \cdot 8 = 16$. (Here $x = 3$.) The answer is (C).

Answer: (C)

22. HARD · DESMOS

The equation $|x + 6| - 7 = \dfrac{x}{3} + c$, where c is a constant, has exactly one solution x. What is the value of $2c + x$?

(A) -11 (B) -6

(C) -5 **(D) -16 ✓**

1. Type `y=abs(x+6)-7` and `y=x/3+c` (accept the slider for c). Tap the corner of the V: $(-6, -7)$.
2. Slide c. The line is gentler than both arms of the V: with c above the right value it cuts the V twice, with c below it misses the V. It cuts exactly once only when it passes through the corner.
3. At the corner $x = -6$ the line has height $-6/3 + c$, which must equal -7. Type `-7-(-6/3)`: it returns -5, so $c = -5$ (the slider shows the line through $(-6, -7)$).
4. The one solution is $x = -6$. Type `2*(-5)+(-6)`: it returns -16. The answer is (D).

Answer: (D)

23. HARD · NO CALCULATOR NEEDED

In the xy-plane, the line $y = mx - 5$ intersects the parabola $y = x^2 - 6x + 11$ at exactly one point, where m is a negative constant. What is the value of m?

(A) 14 **(B) -14 ✓**

(C) -2 (D) 2

1. Set the expressions equal: $x^2 - 6x + 11 = mx - 5$. Move everything to the left: $x^2 - (6+m)x + 16 = 0$.
2. Exactly one intersection point means exactly one solution, so the discriminant is 0: $(6+m)^2 - 4(1)(16) = 0$.
3. So $(6+m)^2 = 64$, giving $6+m = 8$ or $6+m = -8$, i.e. $m = 2$ or $m = -14$.
4. The problem says m is negative, so $m = -14$. The answer is (B). Check: the touch point is $x = \dfrac{6+m}{2} = -4$, and both sides equal 51 there.

Answer: (B)

24. HARD · DESMOS

$$x^2 + y^2 = 65$$

$$x - y = 7$$

The solution (a, b) to the system of equations above satisfies $a > 0$. What is the value of $a + 3b$?

1. Rewrite the second equation as $y = x - 7$. Type `x^2+y^2=65` (a circle) and `y=x-7`.
2. Tap the two points where the line crosses the circle: $(-1, -8)$ and $(8, 1)$. The problem wants $a > 0$.
3. So $(a, b) = (8, 1)$. Type `8+3*1`: it returns 11.

Answer: 11

25. HARD · DESMOS

In the xy-plane, the graphs of $y = x^2 + 6x + 4$ and $y = mx - 4$ have no point in common, where m is a constant. What is the sum of all integer values of m?

1. Set the expressions equal: $x^2 + 6x + 4 = mx - 4$, so $x^2 + (6-m)x + 8 = 0$. No common point means no real solution, so the discriminant must be negative: $(6-m)^2 - 32 < 0$.
2. Type `y=(6-x)^2-32` (letting x stand for m).
3. Tap the two zeros: (0.3431, 0) and (11.6569, 0). The curve is below the axis (negative discriminant) between them, so $0.3431 < m < 11.6569$.
4. The integers in that range are $1, 2, \ldots, 11$. Add them: Type `11*12/2`: it returns 66.

Answer: 66

26. HARD · DESMOS

$$2^x = 3x + k$$

In the equation above, k is a constant. What is the smallest integer value of k for which the equation has two distinct real solutions?

(A) -2 ✓ (B) -1

(C) 0 (D) -3

1. Rewrite the equation as $2^x - 3x = k$. Type `y=2^x-3x` and tap its lowest point: (2.1137, −2.0131).
2. Each value of k is a horizontal line. Type `y=-3`: it is below the lowest point, so it never meets the curve (no solutions).
3. Type `y=-2`: it is just above the lowest point and meets the curve twice, at (2, −2) and (2.2245, −2).
4. Two solutions need $k > -2.0131$, so the smallest integer is -2. The answer is (A).

Answer: (A)

27. HARD · NO CALCULATOR NEEDED

For an integer k, the equation

$$9 - 3\,|2x - 8| = 1 - \frac{5k}{2}$$

has no real solution x. What is the greatest possible value of k?

1. The expression $|2x - 8|$ is never negative, so $9 - 3|2x - 8|$ is at most 9 (it equals 9 when $x = 4$).
2. So the equation has no solution exactly when the right side is bigger than 9: $1 - \frac{5k}{2} > 9$.
3. Subtract 1: $-\frac{5k}{2} > 8$. Multiply by $-\frac{2}{5}$ and flip the inequality: $k < -\frac{16}{5} = -3.2$.
4. The greatest integer below -3.2 is -4. Check: $k = -4$ gives right side $11 > 9$ (no solution), but $k = -3$ gives $\frac{17}{2} \le 9$ (has a solution).

Answer: -4

28. HARD · DESMOS

For a constant k, the equation

$$2\sqrt{x+1} + 2\sqrt{15 - x} = k$$

has exactly one real solution. What is the value of k?

(A) 16 (B) $2 + 2\sqrt{15}$

(C) 8 **(D) $8\sqrt{2}$ ✓**

1. Type `y=2*sqrt(x+1)+2*sqrt(15-x)`. It only exists for $-1 \le x \le 15$. Tap the left end (−1, 8), the top (7, 11.3137) and the right end (15, 8).
2. The solutions are where the horizontal line $y = k$ meets the curve. Type `y=11`: it meets the curve twice, at (3.3621, 11) and (10.6379, 11).
3. Type `y=8`: it meets the curve at both ends, again twice. A line meets the curve exactly once only when it just touches the top.
4. Type `4*sqrt(8)`: it returns 11.3137, which is the top height. Type `8*sqrt(2)`: it returns 11.3137, the same number, so $k = 8\sqrt{2}$. The answer is (D).

Answer: (D)

29. HARD · NO CALCULATOR NEEDED

$$x^2 + y^2 = 45$$
$$y = mx + 15$$

In the system above, m is a negative constant. If the system has exactly one solution (x, y), what is the value of x?

(A) -3 (B) 3

(C) 6 ✓ (D) -6

1. Substitute $y = mx + 15$ into the circle: $x^2 + (mx + 15)^2 = 45$, which expands to $(1 + m^2)x^2 + 30mx + 180 = 0$.
2. Exactly one solution means the discriminant is 0: $(30m)^2 - 4(1 + m^2)(180) = 0$, so $180m^2 - 720 = 0$ and $m^2 = 4$.
3. The problem says m is negative, so $m = -2$. The equation becomes $5x^2 - 60x + 180 = 0$, i.e. $x^2 - 12x + 36 = 0$.
4. That is $(x - 6)^2 = 0$, so $x = 6$. Check: $y = -2(6) + 15 = 3$ and $6^2 + 3^2 = 45$. The answer is (C).
Answer: (C)

30. HARD · NO CALCULATOR NEEDED

$$y = 3x^2 + 4$$
$$y = 1 - kx$$

In the system above, k is a constant. For which of the following values of k does the system have two distinct solutions (x, y) whose x-coordinates are both negative?

(A) 6 **(B) 7 ✓**

(C) -7 (D) 2

1. Set the expressions equal: $3x^2 + 4 = 1 - kx$, so $3x^2 + kx + 3 = 0$.
2. Two distinct solutions need a positive discriminant: $k^2 - 4(3)(3) = k^2 - 36 > 0$, so $k > 6$ or $k < -6$.
3. Both solutions negative: the sum of the solutions is $-\frac{k}{3}$, which must be negative, so $k > 0$. (The product is $\frac{3}{3} = 1 > 0$, so the two solutions already have the same sign.)
4. Together, $k > 6$. Of the options, $k = 7$ works; $k = 6$ gives only one solution, $k = 2$ gives none, and $k = -7$ gives two positive solutions. The answer is (B).
Answer: (B)

31. HARD · DESMOS

For how many integer values of k with $-10 \le k \le 10$ does the equation

$$\sqrt{x+k} = x - 2$$

have exactly one real solution?

1. Square both sides: $x + k = (x-2)^2$, and remember $x - 2 \ge 0$, because a square root is never negative. Solve for k: $k = x^2 - 5x + 4$ with $x \ge 2$.
2. Type `y=x^2-5x+4{x>=2}`. Tap the left end $(2, -2)$ and the lowest point $(2.5, -2.25)$. The number of solutions is the number of times the horizontal line $y = k$ meets this curve.
3. Type `y=-2`: it meets the curve twice, at $(2, -2)$ and $(3, -2)$, so $k = -2$ gives two solutions. Type `y=-1`: it meets the curve only once, at $(3.618, -1)$.
4. Every line from $y = -1$ upward meets the curve exactly once. The values are $k = -1, 0, 1, \ldots, 10$. Count them: Type `10-(-1)+1`: it returns 12. (The single touch at $k = -2.25$ is not an integer.)

Answer: 12

32. HARD · NO CALCULATOR NEEDED

In the xy-plane, the line $y = kx+4$, where k is a constant, is tangent to the parabola $y = x^2+4x+13$ for exactly two values of k. For each of these values the line touches the parabola at one point. What is the greatest y-coordinate of these two points of tangency?

(A) 10 (B) 13

(C) 34 ✓ (D) 3

1. Set the expressions equal: $x^2 + 4x + 13 = kx + 4$. Move everything to the left: $x^2 + (4-k)x + 9 = 0$.
2. Tangent means exactly one solution, so the discriminant is 0: $(4-k)^2 - 4(1)(9) = 0$, so $(4-k)^2 = 36$ and $4 - k = \pm 6$, giving $k = -2$ or $k = 10$.
3. The touching point is the double root $x = -\dfrac{4-k}{2}$: for $k = -2$ it is $x = -3$, and for $k = 10$ it is $x = 3$.
4. Find the y-coordinates with the parabola: $y = (-3)^2 + 4(-3) + 13 = 10$ and $y = 3^2 + 4(3) + 13 = 34$. The greatest is 34. The answer is (C).

Answer: (C)

33. HARD · DESMOS

For a real number k, consider the equation

$$x - 4\sqrt{x} = k.$$

For how many integer values of k from -10 to 10, inclusive, does the equation have exactly one real solution x?

1. Type `y=x-4*sqrt(x)`. Tap the left end $(0, 0)$, the lowest point $(4, -4)$ and the right-hand zero $(16, 0)$. The solutions of the equation are where the horizontal line $y = k$ meets this curve.
2. Type `y=-4`: it only touches the lowest point, so exactly one solution. Type `y=-2`: it meets the curve twice, at $(0.3431, -2)$ and $(11.6569, -2)$. Type `y=0`: twice again, at $(0, 0)$ and $(16, 0)$. Below $y = -4$ there are no solutions.
3. Type `y=3`: only once, at $(21.583, 3)$. Every line above $y = 0$ meets the curve only once, on the rising part to the right of $x = 16$.
4. Exactly one solution happens for $k = -4$ (one value) and for $k = 1, 2, \ldots, 10$ (ten values). Count them: Type `1+10`: it returns 11.

Answer: 11

34. HARD · NO CALCULATOR NEEDED

$$x^3 + (3-k)x^2 - (10+3k)x + 10k = 0$$

In the equation above, k is a constant. The equation has exactly two distinct real solutions for exactly two values of k. What is the sum of these two values of k?

1. Group the terms with k and the terms without: $(x^3 + 3x^2 - 10x) - k(x^2 + 3x - 10) = 0$.
2. Factor each group: $x(x^2 + 3x - 10) - k(x^2 + 3x - 10) = 0$, so $(x - k)(x^2 + 3x - 10) = 0$.
3. Factor the quadratic: $x^2 + 3x - 10 = (x + 5)(x - 2)$. The solutions are $x = -5$, $x = 2$ and $x = k$.
4. There are exactly two distinct solutions only when k repeats one of the fixed solutions: $k = -5$ or $k = 2$. Their sum is -3.

Answer: -3

35. HARD · DESMOS

$$|2x - k| + k = x + 6$$

In the equation above, k is a constant. The equation has exactly one solution. What is the value of k?

1. Type `y=abs(2x-k)+k` and `y=x+6` (accept the slider for k). The first graph is a V whose corner is at $\left(\frac{k}{2},\, k\right)$, and the line $y = x + 6$ is gentler than both arms.
2. Set $k = 0$: the line cuts the V twice, at $(-2,\, 4)$ and $(6,\, 12)$. Slide k up and the two points move together.
3. At $k = 12$ the corner $(6,\, 12)$ sits on the line and there is exactly one intersection. For k larger than that the V is above the line and they never meet.
4. Check with the corner: $\left(\frac{k}{2},\, k\right)$ lies on $y = x + 6$ when $k = \frac{k}{2} + 6$. Type `12/2+6`: it returns 12, which matches $k = 12$.

Answer: 12

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

Emma is 4 years older than Grace. The square of Emma's age, in years, is 64 greater than the square of Grace's age. What is the sum, in years, of Emma's and Grace's ages?

1. Let Emma be e years old and Grace g. We know $e - g = 4$ and $e^2 = g^2 + 64$, so $e^2 - g^2 = 64$.

2. Factor the difference of squares: $(e - g)(e + g) = 64$.

3. Put in $e - g = 4$: $4(e + g) = 64$, so $e + g = 16$.

4. Check: $g = 6$ and $e = 10$ give $10^2 - 6^2 = 100 - 36 = 64$. The sum of their ages is 16.

Answer: 16

37. MEDIUM · NO CALCULATOR NEEDED

As part of a tourism marketing campaign, the number of visitors (in thousands) x weeks after the campaign begins is modeled by $v = x^2 + 2x - 3$. A separate baseline estimate predicts $v = -3$ for the same period. For how many nonnegative values of x are the two estimates equal?

(A) 0

(B) 1 ✓

(C) 2

(D) 3

1. Set the model equal to the baseline: $x^2 + 2x - 3 = -3$.

2. Add 3 to both sides: $x^2 + 2x = 0$.

3. Factor: $x(x + 2) = 0$, so $x = 0$ or $x = -2$.

4. The problem only allows nonnegative x, so $x = -2$ is rejected and only $x = 0$ counts: 1 value. The answer is (B).

Answer: (B)

38. MEDIUM · NO CALCULATOR NEEDED

What is the positive solution to the equation $x^2 - 5x - 24 = 0$?

1. Look for two numbers that multiply to -24 and add to -5: they are -8 and 3.

2. So $x^2 - 5x - 24 = (x + 3)(x - 8)$.

3. Set each factor to 0: $x = -3$ or $x = 8$.

4. The positive solution is 8. Check: $8^2 - 5(8) - 24 = 0$.

Answer: 8

39. MEDIUM · NO CALCULATOR NEEDED

What are the solutions to the equation $2x^2 - 5x - 3 = 0$?

(A) $x = -\frac{1}{2}$ and $x = 3$ ✓

(B) $x = \frac{1}{2}$ and $x = -3$

(C) $x = -\frac{3}{2}$ and $x = 1$

(D) $x = \frac{3}{2}$ and $x = -1$

1. Multiply $a \cdot c = 2 \cdot (-3) = -6$. Find two numbers with product -6 and sum -5: -6 and 1.

2. Split the middle term: $2x^2 - 6x + x - 3 = 0$. Factor by grouping: $2x(x - 3) + 1(x - 3) = (2x + 1)(x - 3) = 0$.

3. Set each factor to 0: $2x + 1 = 0$ gives $x = -\frac{1}{2}$, and $x - 3 = 0$ gives $x = 3$.

4. The solutions are $x = -\frac{1}{2}$ and $x = 3$, which is option (A). Check $x = 3$: $2(9) - 15 - 3 = 0$.

Answer: (A)

40. Medium · No calculator needed

If x and y are positive integers such that $x^2 - y^2 = 45$ and $x - y = 5$, what is the value of $x + y$?

(A) 7 **(B) 9 ✓**

(C) 11 (D) 13

1. Factor the difference of squares: $x^2 - y^2 = (x - y)(x + y)$.
2. Put in $x - y = 5$: $5(x + y) = 45$.
3. Divide by 5: $x + y = 9$, which is option (B).
4. Check: $x + y = 9$ and $x - y = 5$ give $x = 7$ and $y = 2$, both positive integers, and $7^2 - 2^2 = 49 - 4 = 45$.

Answer: (B)

41. Medium · No calculator needed

An integer n is subtracted from its square. Which of the following could NOT be the result?

(A) 0 (B) An even integer

(C) An odd integer ✓ (D) A positive integer

1. Write the result as $n^2 - n = n(n - 1)$: the product of two consecutive integers.
2. Of two consecutive integers one is even, so the product is always even. Try some values: $n = -2$ gives 6, $n = -1$ gives 2, $n = 0$ gives 0, $n = 1$ gives 0, $n = 2$ gives 2, $n = 3$ gives 6.
3. So the result can be 0 (A), an even integer (B) and a positive integer (D), e.g. $n = -2$ gives 6.
4. It can never be odd, so the answer is (C).

Answer: (C)

42. Medium · No calculator needed

For which value of k does the equation $x^2 + 6x + k = 0$ have exactly one real solution?

(A) 3 (B) 6

(C) 9 ✓ (D) 12

1. Exactly one real solution means the discriminant is 0: $b^2 - 4ac = 0$.
2. Here $a = 1$, $b = 6$, $c = k$: $6^2 - 4(1)k = 0$, so $36 - 4k = 0$.
3. Solve: $4k = 36$, so $k = 9$, which is option (C).
4. Check: $x^2 + 6x + 9 = (x + 3)^2$, which equals 0 only at $x = -3$ (one solution).

Answer: (C)

43. Medium · No calculator needed

What is the solution set of the equation $\dfrac{x}{x-2} = \dfrac{2}{x-2} + 3$?

(A) $x = 2$ only

(B) $x = 4$ only

(C) $x = 2$ and $x = 4$

(D) There is no solution. ✓

1. Multiply every term by $x - 2$ (the common denominator, with $x \neq 2$): $x = 2 + 3(x - 2)$.
2. Expand: $x = 2 + 3x - 6$, so $x = 3x - 4$.
3. Subtract $3x$: $-2x = -4$, so $x = 2$.
4. But $x = 2$ makes the denominator $x - 2$ equal to 0, so it is not allowed (an extraneous solution). There is no solution, which is option (D).

Answer: (D)

44. Medium · No calculator needed

Which inequality is equivalent to $8x - 12y > 20$?

(A) $2x - 3y > 5$ ✓

(B) $3x - 2y > 5$

(C) $4x - 3y > 5$

(D) $2y - 3x > 5$

1. Divide every term of $8x - 12y > 20$ by the same positive number 4.
2. The inequality sign stays the same because $4 > 0$: $\dfrac{8x}{4} - \dfrac{12y}{4} > \dfrac{20}{4}$.
3. That gives $2x - 3y > 5$, which is option (A).
4. Check with test points. At $(4, 0)$ both are true: $8(4) - 12(0) = 32 > 20$ and $2(4) - 3(0) = 8 > 5$. At $(2, 0)$ both are false: $16 > 20$ and $4 > 5$ are false. But $3x - 2y > 5$ is true at $(2, 0)$ since $6 > 5$, so it is not equivalent.

Answer: (A)

45. Medium · No calculator needed

How many distinct real solutions does the equation $4x^2 - 20x + 25 = 0$ have?

(A) Exactly one ✓

(B) Exactly two

(C) Infinitely many

(D) Zero

1. Compute the discriminant: $b^2 - 4ac = (-20)^2 - 4(4)(25) = 400 - 400 = 0$.
2. A discriminant of 0 means exactly one real solution.
3. Check by factoring: $4x^2 - 20x + 25 = (2x - 5)^2$, so $x = \dfrac{5}{2}$ is the only solution.
4. The answer is (A).

Answer: (A)

46. Medium · No calculator needed

In the system of equations below, x and y satisfy both equations.

$$y = 5x + 2$$

$$y = 5x^2 + 2$$

Which ordered pair (x, y) is a solution to this system?

(A) $(0, 0)$ (B) $(0, 5)$

(C) $(1, 5)$ **(D)** $(1, 7)$ ✓

1. Both equations equal y, so set the right sides equal: $5x + 2 = 5x^2 + 2$.
2. Subtract $5x + 2$ from both sides: $0 = 5x^2 - 5x = 5x(x - 1)$, so $x = 0$ or $x = 1$.
3. Find y from $y = 5x + 2$: $x = 0$ gives $y = 2$ and $x = 1$ gives $y = 7$. The solutions are $(0, 2)$ and $(1, 7)$.
4. Only $(1, 7)$ is listed, which is option (D). The others fail: $(0, 0)$ and $(0, 5)$ have $y \neq 2$ at $x = 0$, and $(1, 5)$ has $y \neq 7$ at $x = 1$.

Answer: (D)

47. Medium · No calculator needed

$$7|x| = 63$$

What is the positive solution to the given equation?

1. Divide both sides by 7: $|x| = \dfrac{63}{7} = 9$.
2. The absolute value equation has two solutions: $x = 9$ and $x = -9$.
3. The question asks for the positive one: $x = 9$.
4. Check: $7|9| = 63$.

Answer: 9

48. Medium · No calculator needed

What is the sum of the y-coordinates of the solutions to the system below?

$$y = x^2 - 4x$$

$$y = 2x - 5$$

(A) -2 (B) 0

(C) 2 ✓ (D) 4

1. Set the expressions equal: $x^2 - 4x = 2x - 5$. Move everything to the left: $x^2 - 6x + 5 = 0$.
2. Factor: $(x - 1)(x - 5) = 0$, so $x = 1$ or $x = 5$.
3. Find each y from the line $y = 2x - 5$: $x = 1$ gives $y = -3$, and $x = 5$ gives $y = 5$.
4. The sum of the y-coordinates is $-3 + 5 = 2$, which is option (C).

Answer: (C)

49. Medium · No calculator needed

The height of a triangle is 4 cm more than its base. If the area of the triangle is 30 cm^2, what is the length of the base, in cm?

(A) 4 (B) 5

(C) 6 ✓ (D) 8

1. Let the base be b cm. The height is $b+4$ cm. The area of a triangle is $\frac{1}{2} \cdot \text{base} \cdot \text{height}$, so $\frac{1}{2}b(b+4) = 30$.

2. Multiply by 2: $b^2 + 4b = 60$, so $b^2 + 4b - 60 = 0$.

3. Factor: $(b+10)(b-6) = 0$, so $b = -10$ or $b = 6$.

4. A length is positive, so $b = 6$ cm, which is option (C). Check: $\frac{1}{2}(6)(10) = 30$.

Answer: (C)

50. Medium · No calculator needed

What is the solution to the equation $\dfrac{x+5}{x-1} = 3$, for $x \neq 1$?

1. Multiply both sides by $x-1$ (allowed since $x \neq 1$): $x + 5 = 3(x-1)$.

2. Expand: $x + 5 = 3x - 3$.

3. Move the x terms to one side and numbers to the other: $5 + 3 = 3x - x$, so $8 = 2x$.

4. Divide by 2: $x = 4$. Check: $\dfrac{4+5}{4-1} = \dfrac{9}{3} = 3$.

Answer: 4

51. Medium · No calculator needed

The curve $y = \dfrac{x^2}{3}$ and the line $y = \dfrac{2x}{3}$ intersect at the origin and at the point (a, b). What is the value of b?

1. Set the expressions equal: $\dfrac{x^2}{3} = \dfrac{2x}{3}$. Multiply by 3: $x^2 = 2x$.

2. Move everything to one side and factor: $x^2 - 2x = x(x-2) = 0$, so $x = 0$ or $x = 2$.

3. The origin is the point with $x = 0$, so the other point has $a = 2$.

4. Then $b = \dfrac{2a}{3} = \dfrac{2 \cdot 2}{3} = \dfrac{4}{3}$. Check with the parabola: $\dfrac{2^2}{3} = \dfrac{4}{3}$.

Answer: $\dfrac{4}{3}$

52. Medium · No calculator needed

For what values of x is the expression $2x^2 - 2x - 24$ equal to 0?

(A) $x = -4,\ x = 3$ **(B) $x = 4,\ x = -3$ ✓**

(C) $x = 4,\ x = 3$ (D) $x = -4,\ x = -3$

1. Divide every term by 2: $x^2 - x - 12 = 0$.

2. Find two numbers with product -12 and sum -1: -4 and 3. So $x^2 - x - 12 = (x-4)(x+3)$.

3. Set each factor to 0: $x = 4$ or $x = -3$.

4. That is option (B). Check $x = -3$: $2(-3)^2 - 2(-3) - 24 = 0$.

Answer: (B)

53. Medium · No calculator needed

If $(x-3)(x+7) = -25$, what is the value of x?

1. Expand the left side: $(x-3)(x+7) = x^2 + 4x - 21$. So $x^2 + 4x - 21 = -25$.
2. Add 25 to both sides: $x^2 + 4x + 4 = 0$.
3. Factor: $(x+2)^2 = 0$, so $x = -2$.
4. Check: $(-2-3)(-2+7) = (-5)(5) = -25$.

Answer: -2

54. Medium · No calculator needed

The square of a positive number is 12 more than 4 times the number. What is the number?

1. Translate: the square of the number is 12 more than 4 times the number, so $x^2 = 4x + 12$.
2. Move everything to the left side: $x^2 - 4x - 12 = 0$.
3. Factor: $(x+2)(x-6) = 0$, so $x = -2$ or $x = 6$.
4. The number is positive, so it is 6. Check: $6^2 = 36$ and $4(6) + 12 = 36$.

Answer: 6

55. Medium · No calculator needed

$$xy = 48$$
$$3x - y = 0$$

If (x, y) is a solution to the system above and $x > 0$, what is the value of y?

1. From $3x - y = 0$ we get $y = 3x$.
2. Substitute into $xy = 48$: $x(3x) = 48$, so $3x^2 = 48$.
3. Divide by 3: $x^2 = 16$, so $x = \pm 4$. We need $x > 0$, so $x = 4$.
4. Then $y = 3x = 12$. Check: $xy = 4 \cdot 12 = 48$.

Answer: 12

56. Medium · No calculator needed

If $2^{x+3} - 2^x = 56$, what is the value of x?

1. Rewrite 2^{x+3} as $2^x \cdot 2^3 = 8 \cdot 2^x$. The equation becomes $8 \cdot 2^x - 2^x = 56$.
2. Combine like terms: $7 \cdot 2^x = 56$.
3. Divide by 7: $2^x = 8$.
4. Since $8 = 2^3$, $x = 3$. Check: $2^6 - 2^3 = 64 - 8 = 56$.

Answer: 3

57. Medium · No calculator needed

What is the product of the positive solutions to the equation $x^4 - 13x^2 + 36 = 0$?

1. Let $u = x^2$. Then $x^4 = u^2$ and the equation becomes $u^2 - 13u + 36 = 0$.

2. Factor: $(u - 4)(u - 9) = 0$, so $u = 4$ or $u = 9$.

3. Since $u = x^2$: $x^2 = 4$ gives $x = \pm 2$, and $x^2 = 9$ gives $x = \pm 3$. The positive solutions are 2 and 3.

4. Their product is $2 \cdot 3 = 6$.

Answer: 6